Final Project

Goals

The final project is designed to:

1. Strengthen your understanding of the concepts covered in class.
2. Expose you to state-of-the-art research in software security.
3. Provide hands-on experience applying what you have learned.
4. Enhance teamwork, collaboration, and communication skills.
5. Improve your technical presentation skills.

Candidate Project Formats

Each group will select **one** of the following project formats:

1. **Survey-Based**
   - 1.1. Choose a research topic that we discussed in class.
   - 1.2. Select at least 5 papers on this topic, published in **top-tier conferences within the past three years** (see the list of conferences below).
   - 1.3. Write a survey report of at least 10 pages (double-column, 10pt font). The report should synthesize the findings, highlight similarities/differences across papers, and include your critical analysis of the state of research.
2. **Programming-Based**

   You may choose **one** from the examples below or propose your own:
   a. Develop a biometric authentication tool (e.g., fingerprint or face recognition).
   b. Develop a tool to securely store and verify user passwords (with unique salts and slow hashing). You must write your own code; do not use ready-made libraries like bcrypt.
   c. Build a symbolic execution tool that can discover bugs or vulnerabilities in programs.
   d. Propose your own project idea. For approval, email me a short description of your idea and its goals. You may begin only after your proposed project is approved.

3. **Hybrid**
   3.1. Select a topic covered in class and choose one recent paper on this topic, published in **a top-tier conference within the past three years** (see the list of conferences below). Write a 2-page report (double-column, 10pt font) that summarizes the paper and demonstrates your critical thinking.
   3.2. **In addition**, choose **one** of the following:
      a. Re-implement the system or algorithm described in the paper. You are encouraged to make use of any available open-source artifacts associated with the paper.
      b. Complete one project from the Software Security Labs category on the SEED Security Labs platform.

Project Selection Deadline

Please send me an email indicating which format your group has chosen by **October 19**.

- If you later decide your chosen format is not a good fit, you may switch to another option after October 19, but please confirm with me.

Submission Requirements

Each group must submit the following:

1. **Final Project Report**
   1.1 Survey-Based Projects
   At least 10 pages, double column, 10pt font. The report must demonstrate critical analysis, not just summaries.
   1.2 Programming-Based Projects:
   No fixed page requirement. The report should include the system architecture/workflow, identify which components you implemented, and highlight any reused code or frameworks. Source code must also be submitted.
   1.3 Hybrid Projects
   - **Paper report**: at least 2 pages, double column, 10pt font, with critical reflection.

- **Hands-on component**: either meet the requirements defined by the SEED Security Labs or provide documentation for your re-implementation of the chosen paper.

2. **Presentation/Demo**
   - Each group will give an in-class presentation or demo of their project.
   - You must also submit your presentation/demo slides.
3. **Submission Deadline**
   All final project materials are due by **December 7 at 11:59 PM**.

List of Conferences

Abbreviation	Full Name
S&P	IEEE Symposium on Security and Privacy
CCS	ACM Conference on Computer and Communications Security
USENIX Security	Usenix Security Symposium
NDSS	ISOC Network and Distributed System Security Symposium
CRYPTO	International Cryptology Conference
EUROCRYPT	European Cryptology Conference
FSE	Fast Software Encryption
ICSE	International Conference on Software Engineering
ASE	International Conference on Automated Software Engineering